

Empirical Research in Finance

9. Event studies and Difference-in-Differences

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Today's roadmap

1. Why these tools matter
2. Event studies: AR, CAR, tests
3. Difference-in-Differences: logic, plot, regression
4. Combining ES and DiD

Notation we will use today

$R_{i,t}$	Return of stock i on day t
$R_{m,t}$	Market return on day t
$E[R_{i,t} \mathbf{x}_t]$	Expected (“normal”) return absent the event
$AR_{i,t}$	Abnormal return: actual minus expected
$\overline{AR}_t$	Average AR across firms on day t
$CAR(t_1, t_2)$	Cumulative AR from t_1 to t_2
τ	Event date for a given firm
$Treated_i$	1 if firm/unit is exposed to the policy
$Post_t$	1 if period t is after the policy
δ	DiD coefficient

Finance is filled with events

Examples:

- ▶ Quarterly earnings announcements
- ▶ Dividend cuts and increases
- ▶ Mergers, acquisitions, SEOs
- ▶ Credit-rating changes
- ▶ Regulatory shocks

The two questions empirical finance keeps asking:

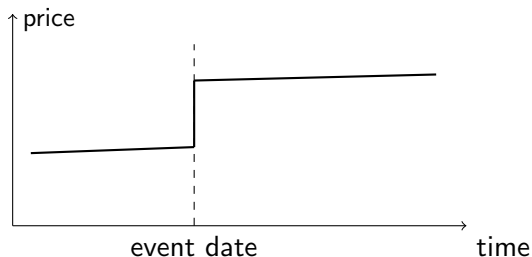
1. How do prices react to news? → **Event Studies**
2. Did this event change firm behaviour? → **DiD**

Both rely on the same idea: *counterfactuals*.

Event studies

Event studies measure the impact of a recurring type of event on security prices.

Key idea: If markets are informationally efficient, the information content of the event should affect prices around the announcement.



Outline of an event study

1. Compute *adjusted - or abnormal - returns*

$$AR_{i,t} := R_{i,t} - E[R_{i,t}|\mathbf{x}_t]$$

where

- ▶ $R_{i,t}$: return of firm i at (or around) event date t ;
 - ▶ $E[R_{i,t}|\mathbf{x}_t]$: model return that does *not* take the event into account.
2. Then compute averages $\overline{AR}$ across many events. Does it jump?

Models for predicted returns

Default model, the **market model**:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$$

Estimate $(\hat{\alpha}_i, \hat{\beta}_i)$ by OLS on a pre-event window. Predicted return $= \hat{\alpha}_i + \hat{\beta}_i R_{m,t}$.

Other choices, when needed:

- ▶ Constant mean-return model: $R_{i,t} = \mu_i + \varepsilon_{i,t}$ (often too simplistic).
- ▶ Multi-factor model (e.g. Fama-French):
 $R_{i,t} = \alpha_i + \beta_{m,i} R_{m,t} + \beta_{s,i} \text{SMB}_t + \beta_{h,i} \text{HML}_t + \varepsilon_{i,t}$.
- ▶ Market-adjusted return (no estimation window, e.g. IPOs):
 $R_{i,t} = R_{m,t} + \varepsilon_{i,t}$.

Over short windows the simple market model and FF3 usually give very similar abnormal returns; choose for **economic** reasons, not mechanical.

Event and estimation windows

Market prices may react not only at the time of the event, but also

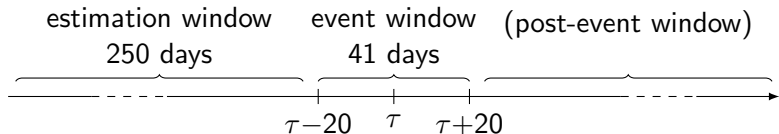
- ▶ *slightly before*, e.g. because of insider trading, or
- ▶ *slightly after*, e.g. if information takes time to diffuse.

⇒ look at an *event window* around the event.

Model parameters need to be estimated on a clean window.

- ▶ The model must reflect returns *in the absence* of the event
⇒ pick an estimation window that is *not* in the event window.

Widen the window for leakage and diffusion, but no further — extra days only add noise.



Aggregation — from AR to CAR

- Aggregate across N securities/firms:

$$\overline{AR}_t = \frac{1}{N} \sum_{i=1}^N \widehat{AR}_{i,t}$$

- Aggregate across time, from t_1 to t_2 , to compute the cumulative average abnormal return:

$$CAR(t_1, t_2) = \sum_{t=t_1}^{t_2} \overline{AR}_t$$

CAR shows the total effect of the event over the chosen window.

Worked example — one firm, 5-day window

TechCo announces very positive earnings on day 0. Fit the market model on the **estimation window** $\implies \hat{\alpha} = 0.0001, \hat{\beta} = 1.10$.

Day t	$R_{i,t}$	$R_{m,t}$	Pred = $\hat{\alpha} + \hat{\beta}R_m$	$\widehat{AR}_{i,t}$	CAR
-2	+0.40%	+0.30%	+0.34%	+0.06%	+0.06%
-1	+0.20%	+0.10%	+0.12%	+0.08%	+0.14%
0	+ 1.80%	+ 0.20%	+ 0.23%	+ 1.57%	+ 1.71%
+1	+0.60%	+0.40%	+0.45%	+0.15%	+1.86%
+2	+0.10%	+0.20%	+0.23%	-0.13%	+1.73%

- ▶ $CAR(-2, +2) = +1.73\%$.
- ▶ Most of the action is on day 0; small reversal on day +2.

From one firm to many — average AR and CAR

Now suppose **three** firms announce earnings, each with its own abnormal returns. Average across firms each day, then cumulate:

Day t	-2	-1	0	+1	+2
TechCo	+0.06	+0.08	+1.57	+0.15	-0.13
Firm B	+0.10	+0.02	+1.30	+0.16	+0.07
Firm C	-0.04	+0.08	+1.03	+0.05	-0.06
$\overline{AR}_t$	+0.04	+0.06	+1.30	+0.12	-0.04
CAR (cumul.)	+0.04	+0.10	+1.40	+1.52	+1.48

(all numbers in %)

- ▶ $\overline{AR}_t = \frac{1}{3} \sum_i \widehat{AR}_{i,t}$: firm-specific variations partly cancel, the day-0 jump survives.
- ▶ $CAR(-2, +2) = +1.48\%$ — the average total reaction across the three firms.

Averaging isolates the common signal from idiosyncratic noise — with more firms, the cleaner the picture.

Tests — is the reaction real, or just noise?

- ▶ $H_0 : \overline{AR}_t = 0: \quad t = \frac{\overline{AR}_t}{SE(\overline{AR})}$
- ▶ $H_0 : CAR = 0: \quad t = \frac{CAR(t_1, t_2)}{SE(CAR)}$

Each test compares the **size of the reaction** to its **standard error** — the stock's normal daily variation.

Where does that come from? It is the **standard deviation of the average abnormal return** — averaged across firms each day — over the estimation window *before* the event.

Rule of thumb: $|t| > 1.96 \Rightarrow$ statistically significant.

Example: on day 0, $\overline{AR}_0 = 1.30\%$ and $\sigma(\overline{AR}) = 0.30\%$, so $t = 1.30/0.30 \approx 4.3 \Rightarrow$ reject H_0 .

Tests — CARS

For a CAR over several days, the daily noise piles up — **multiply the SE by $\sqrt{\text{number of days}}$** :

$$\text{SE}(\text{CAR}) = \text{SE}(\overline{AR}) \times \sqrt{\text{days}}.$$

For a 5-day window: $0.30\% \times \sqrt{5} \approx 0.67\%$. (This assumes the daily abnormal returns are **independent**.)

Tests — CARs over different windows

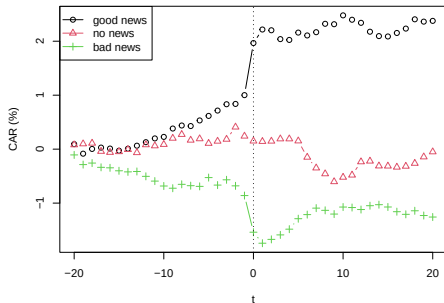
Back to our three firms ($SE(\overline{AR}) = 0.30\%$). Cumulate the average abnormal returns over each window and test $H_0: CAR = 0$ with $t = CAR/SE(CAR)$, where $SE(CAR) = 0.30\% \times \sqrt{L}$:

Window	L	CAR	$SE = 0.30\sqrt{L}$	$t = CAR/SE$
(0, 0)	1	+1.30%	0.30%	+4.33
(-1, +1)	3	+1.48%	0.52%	+2.85
(-2, +2)	5	+1.48%	0.67%	+2.21

- ▶ All three $|t| > 1.96 \Rightarrow$ **significant reaction** in every window.
- ▶ Widening the window adds noise here (the news is concentrated on day 0), so the t -stat **falls** as the window grows.

MacKinlay — 600 real events

600 real earnings announcements (30 DJIA firms, 1989–1993), split by the size of the earnings surprise into **good** / **no** / **bad** news.



- ▶ **Sign of the news, sign of the reaction:** good drifts up, bad drifts down, no news flat — markets price the information.
- ▶ **More events, sharper result:** averaging over hundreds of events cancels the noise, so even a small reaction shows up as clearly significant — far more than for a single firm.

What can you do with a CAR/AR? Use it as an outcome

Each event gives you **one number per firm**: its *CAR* (or, for a single day, its *AR*).

That number can be the **dependent variable** in a regression — so you can ask *why* some firms react more than others:

$$CAR_i = a + b \text{ Surprise}_i + c \text{ Size}_i + e_i.$$

$b > 0$: the bigger the earnings surprise, the bigger the price jump;
 c tells you whether larger firms react differently. Add more firm features — leverage, governance — as extra regressors.

The event study gives you the outcome; the regression explains it.

Doing this at scale — let a package do it

With many events you use a dedicated package – lines up every firm in **event time** (each firm's τ becomes day 0) and runs the estimation, windows, aggregation and tests.

R, package *eventstudies* (schematic):

```
es <- eventstudy(firm.returns = ret_matrix,  
                 event.list   = events_df,  
                 event.window = 20,  
                 type         = "marketModel",  
                 model.args   = list(market.returns = mkt_ret))  
summary(es); plot(es)
```

Stata: *eventstudy2* (ssc install eventstudy2). Python: no standard package — build it with *statsmodels*.

Common pitfalls

- ▶ **Another big event in the window.** A macro surprise on the same day contaminates the abnormal return.
Fix: keep the window short.
- ▶ **The event leaks into the estimation window.** The “normal” model is no longer clean.
Fix: leave a gap before the event.
- ▶ **Many firms share the same event date.** Their reactions are not independent, so the t -stat is overstated.
Fix: cluster standard errors by date.

When an event study is not enough

Event studies tell us:

- ▶ what investors *expected* would happen,
- ▶ within a few days of the announcement,
- ▶ using asset-pricing models for the counterfactual.

Limit: investor expectations $\neq$ actual real outcomes. Did employment really change? Did lending really fall? Did investment really rise?

DiD picks up where ES stops:

- ▶ studies medium- and long-run real outcomes;
- ▶ uses a **control group** as the counterfactual;
- ▶ compares the change in treated firms to the change in similar untreated firms.

DiD — when and why

Use DiD when you have:

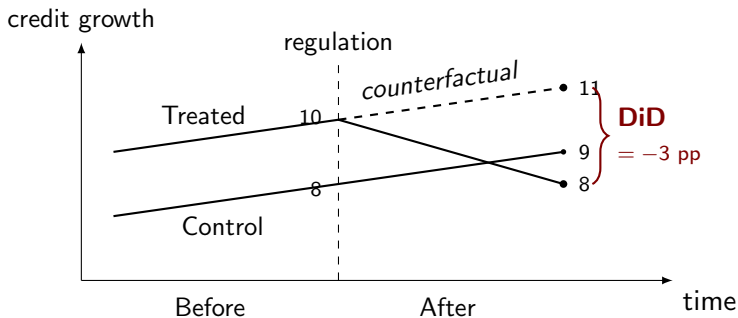
- ▶ a policy or shock that affects some firms/regions/banks but not others;
- ▶ outcomes you can observe before *and* after, for both groups;
- ▶ a credible reason to believe the two groups would have moved in parallel without the policy.

Example . A new regulation tightens leverage limits for banks in State A but not in State B. We want the causal effect of the regulation on credit growth.

- ▶ Treated banks = State A. Control banks = State B.
- ▶ Outcome = credit growth. Time = before vs after the rule.

Graphical intuition — parallel trends

State A banks (treated) and State B banks (control) trend in parallel before the regulation. The control group is unaffected — its line just continues. Treated falls to 8%; its **counterfactual** — treated shifted by the **control group's** change — would have been 11%.



The control group builds the counterfactual; the gap between treated and counterfactual is the $DiD = \Delta Treated - \Delta Control = -3 \text{ pp}$.

Compute DiD by hand — the 2×2 table

Bank credit growth (%):

	Before	After
Treated (State A)	10%	8%
Control (State B)	8%	9%

$$\Delta \text{Treated} = 8 - 10 = -2 \text{ pp}$$

$$\Delta \text{Control} = 9 - 8 = +1 \text{ pp}$$

$$\text{DiD} = (-2) - (+1) = -3 \text{ pp}$$

Interpretation: the control group grew +1 pp anyway, so treated credit growth ended up 3 percentage points lower than it would have without the regulation.

DiD as a regression

$$Y_{i,t} = \alpha + \beta \text{Treated}_i + \gamma \text{Post}_t + \delta (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{i,t}$$

- ▶ α : baseline level (before).
- ▶ β : permanent gap between treated and control.
- ▶ γ : common time trend (affects everyone).
- ▶ δ : DiD effect — the estimate we care about.

$\hat{\delta}$ from this regression equals the -3 pp on the previous slide.

Combining the two — shareholder activism

Event study:

- ▶ Event = activist files 13D ($\geq 5\%$ stake disclosed).
- ▶ Window = ± 5 days.
- ▶ Compute CAR for the targeted firm.
- ▶ Question: do markets expect activism to create value?

Typical finding: $CAR \approx +6-7\%$ in a 5-day window.

DiD complement:

- ▶ Treated = targeted firms.
- ▶ Control = matched firms (size, industry, pre-trend).
- ▶ Outcomes: ROA, Tobin's Q, payout, leverage, investment.
- ▶ Question: do real outcomes actually improve over 2–3 years?

Key takeaways and reading

- ▶ Event studies measure short-run market reactions.
- ▶ DiD identifies (causal) effects of policies and shocks over time.
- ▶ Both rely on credible counterfactuals.
- ▶ Both are widely used in modern empirical finance.

Reading:

- ▶ MacKinlay, A.C., 1997. “Event studies in economics and finance.” *Journal of Economic Literature*, 35(1), pp. 13–39.
- ▶ Brav, Jiang, Partnoy and Thomas, 2008. “Hedge fund activism, corporate governance, and firm performance.” *Journal of Finance*, 63(4), pp. 1729–1775.